Modeling of Zernike Optical Aberrations by MTF and PSF

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Abstract: Currently there is considerable interest in correcting the higher-order optical aberrations of the human eye [1]. Present-day laser eye surgery techniques like PRK and LASIK correct only defocus and astigmatism, which are 2nd order aberrations. Surgically correcting the additional higher-order aberrations could extend the limits of human vision beyond 20/20. In addition, this type of capability could be used to eliminate the higher-order aberrations that may have been caused by a prior surgical procedure. These induced aberrations are responsible for the glare, halos, and other visual defects that have been reported by some eye surgery patients.

Introduction:

The main application involves the high-resolution imaging of the retina and other structures (rods, cones, nerve fibers, blood vessels, capillaries, pigment cells, etc.,) using adaptive optics for the diagnosis and treatment of eye disorders [1].

To determine and apply the proper correction that will eliminate or minimize these aberrations, their characteristics must first be captured and quantitatively described. The wave aberration function suits this purpose well because it completely describes the aggregate effects of the cornea and lens on light passing through every location in the pupil. It defines how the phase of the light is affected after it has propagated through the optical system. In the case of laser eye surgery, if you know the wave aberration function, you know what must be done to the refractive properties of the eye to create an image on the retina that is limited only by diffraction and scattering. For retinal

imaging with adaptive optics, if you know the wave aberration function, you know what must be done to the light entering the pupil to

create an image on the retina that is limited only by diffraction and scattering [2].

Understanding what is needed to correct defocus and astigmatism can be accomplished by having patients view targets through sets of sphero-cylinder test lenses. However, determining the other higher order wave aberrations requires more sophisticated techniques. Wavefront measuring instruments like the Shack-Hartmann Sensor are being utilized to provide data from which the wave aberrations of the human eye can be accurately and reliably estimated.

For the above reasons a mathematical formulation is needed, not only to quantitatively describe the aberrations, but also to facilitate accurate estimation of the wave aberration function from measurement data. In response to the perceived need in the vision community, the Optical Society of America formed a task force and charged it with the mission of developing recommendations on definitions, conventions, and standards for the reporting of optical aberrations of human eyes. One major outcome has been a recommendation adopting the use of Zernike Polynomials for this purpose [3]. The objectives of this course project include an explanation of why Zernike polynomials are preferred over other functions, a mathematical definition of Zernike polynomials, their use in describing

the wave aberration function, a simulation showing the effects of the higher order aberrations on image quality, and an illustration of how Zernike polynomials are used to estimate the wave aberration function from measurement data.

The Effects of Aberrations on Image Quality

Aberrations negatively impact image quality. They change the size and shape of impulse response or point spread function (PSF), which blurs the image. In terms of frequency analysis, the frequency response of the optical system is reduced by phase distortion within the passband.

The effects of aberrations can therefore be characterized by calculating the PSF or MTF (Modulation Transfer Function) of the optical system.

Calculating the PSF and MTF

The image of a point object formed by the optical system is the point spread function or impulse response. It is defined as :

$$PSF(x.y) = \frac{1}{A^2 d^2 A_P}\left\| FT\left(P(x.y).e^{-i\frac{2\pi}{\lambda}\omega(x,y)}\right)_{f_{x=\frac{x}{\lambda d}}, f_{x=\frac{y}{\lambda d}}}\right\| 2 \quad (1)$$

The Modulation Transfer Function is modulus of the Optical Transfer Function. The OTF and MTF are given by :

$$OTF(S_x, S_y) = \frac{FT\{PSF\}}{FT\{PSF\}S_x = S_y = 0} = \frac{FT\left\|FT\left(P(x.y).e^{-i\frac{2\pi}{\lambda}\omega(x,y)}\right)_{f_{x=\frac{x}{\lambda d}}, f_{x=\frac{y}{\lambda d}}}\right\| 2}{FT\left\|FT\left(P(x.y).e^{-i\frac{2\pi}{\lambda}\omega(x,y)}\right)_{f_{x=\frac{x}{\lambda d}}, f_{x=\frac{y}{\lambda d}}}\right\| 2\, S_x = S_y = 0} \quad (2)$$

$$MTF(S_x, S_y) = \|OTF(S_x, S_y)\| \quad (3)$$

The OTF can also be calculated as the auto-correlation of the generalized pupil function, but because the FFT will be used for computational efficiency, the OTF will be calculated using the Fourier transform of the PSF.

For visual applications it is often more convenient to express the PSF and MTF in terms of visual angle. These expressions are given by [6]:

$$PSF(sin(\theta_x), Sin(\theta_y) = \frac{\lambda^2}{A_p}\left\|FT\left(P(x/\lambda.y/\lambda).e^{-i\frac{2\pi}{\lambda}\omega(x/\lambda, y/\lambda)}\right)\right\| 2 \quad (4)$$

$$PSF(\theta_x), (\theta_y) \cong \frac{\lambda^2}{A_p}\left\|FT\left(P(x/\lambda.y/\lambda).e^{-i\frac{2\pi}{\lambda}\omega(x/\lambda, y/\lambda)}\right)\right\| 2 \quad (5)$$

(The eq.5 use for small angles.)

$$OTF(S_x, S_y) = \frac{FT\{PSF(\theta_x),(\theta_y)\}}{FT\{PSF(\theta_x),(\theta_y)\}\ \ |S_x = S_y = 0} \quad (6)$$

$$MTF(S_x, S_y) = \|OTF(S_x, S_y)\| \quad (7)$$

The calculations and computational domains are illustrated in Figures 1 and 2.

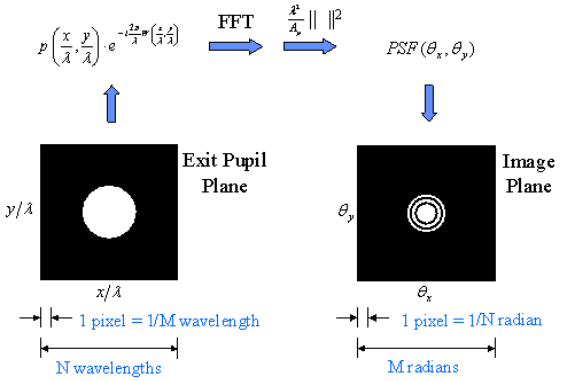


Fig1. PSF calculation and Computational Domains.

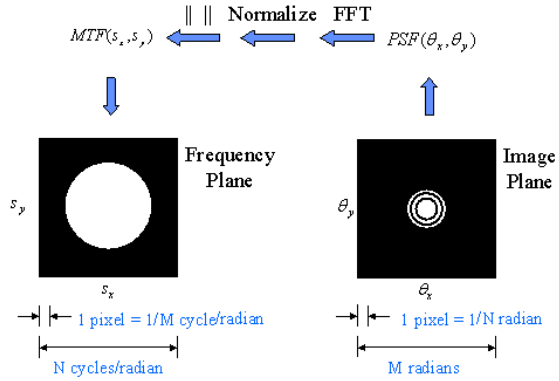


Figure 2. MTF Calculation and Computational Domains.

Results and Discussion

Two Matlab functions, factorial.m and zernike.m were written to generate values for any Zernike polynomial specified by n, m, x, y, and d. Where x and y are 1-D arrays of the x-coordinate and y-coordinate pupil values, and d is the pupil diameter[5]. A Matlab script, ZernikePolynomial.m, was created to generate an image and surface plot of any Zernike polynomial specified by n, m, x, y, and d. Figure 3 contains the images of Zernike polynomials, generated by ZernikePolynomial.m, up to 6th order in a pyramid arrangement.

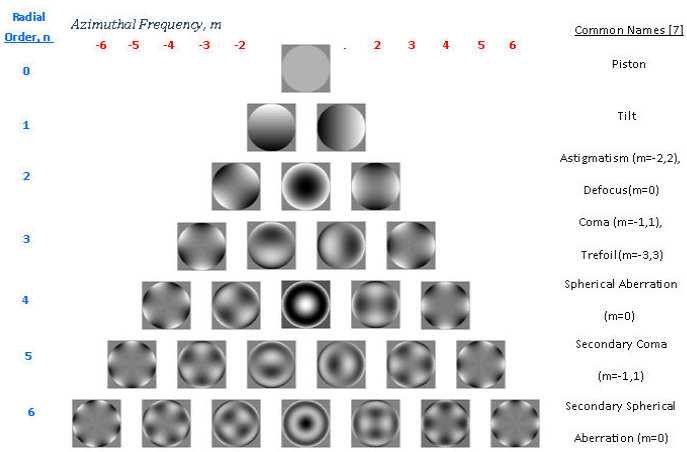
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Two Matlab scripts, ZernikePolynomialPSF.m and ZernikePolynomialMTF.m utilize equations 4-7 to generate and plot the PSF and MTF for any wave aberration mode associated with any Zernike polynomial specified by n, m, pupil diameter (3 to 8 mm), RMS wavefront error (in microns), and wavelength (in nm). These scripts were used to generate the corresponding PSF and MTF images and plots for the wave aberration modes up to 6th order [3] corresponding to each of the Zernike polynomials in Figure 4.

Figure 4 shows the PSF images for a 4 mm pupil diameter at λ = 570 nm and an RMS wavefront error equal to 0.3 micron = 0.53 λ. The PSFs were scaled to achieve saturation to make the fine spatial structure [4] more visible in the images.

Figures 5 plot the MTFs, up to 6th order for 4 mm and 7.3 mm pupil diameters at λ = 570 nm, and an RMS wavefront error equal to 0.114 micron = 0.2 λ . An RMS wavefront error of 0.07 λ or lower is considered diffraction limited, i.e. the aberration is small enough so effectively the light is only affected by diffraction.

Figure 3. Zernike Polynomials up to 6th Order



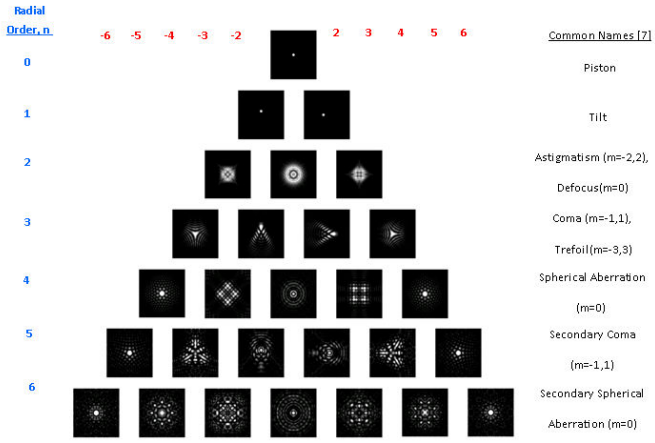


Figure 4. The aberration of eye in order of 6th and R= 4mm Pupil Diameter .

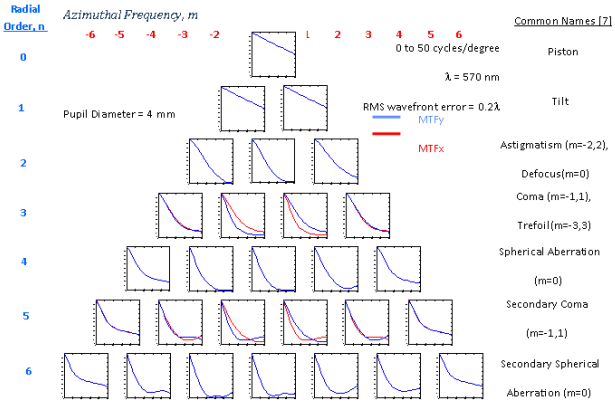


Figure 5. MTFs of of Wave Aberrations Corresponding to Each Zernike Mode up to 6th Order,

4 mm Pupil Diameter, RMS Wavefront Error = 0.2 λ

References

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